

7/24/23, 3:00 PM

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Filetree

SANDIPAN DEY

Third-Party Packages

Third-Party Packages (master)

example1

settings

Terminal

```
codio@athenlight-nevadalopez:~/workspace$ cd example1
codio@athenlight-nevadalopez:~/workspace/example1$ source example-venv/bin/activate
-bash: example-venv/bin/activate: No such file or directory
codio@athenlight-nevadalopez:~/workspace/example1$ python3 -m pip install beautifulsoup4
Collecting beautifulsoup4
  Downloading https://files.pythonhosted.org/packages/57/f4/a69c28ee4f66001a7ded1ac57f29be9378e04edfcb9c5216b923d4bebc/b
beautifulsoup4-4.12.2-py3-none-any.whl (142kB)
    100% |#####| 143kB 5.1MB/s
Collecting soupsieve>1.2 (from beautifulsoup4)
  Downloading https://files.pythonhosted.org/packages/16/e3/4ad79882b92617e3a4a8df1960d6bce08edfb637737ac5c3f3ba29022e25/d
soupsieve-2.3.2.post1-py3-none-any.whl
Installing collected packages: soupsieve, beautifulsoup4
Successfully installed beautifulsoup4-4.12.2 soupsieve-2.3.2.post1
codio@athenlight-nevadalopez:~/workspace/example1$ python3 -m pip list
WARNING: The default format will switch to columns in the future. You can use --format=(legacy|columns) (or define a f
ormat=(legacy|columns) in your pip.conf under the [list] section) to disable this warning.
Package            Version
-----            -
asgiref            3.4.1
asn1crypto         0.24.0
beautifulsoup4     4.12.2
certifi            2018.1.18
chardet            3.0.4
cryptography       2.1.4
decorator          4.1.2
django             3.2.5
idna               2.8
ipython            5.5.0
ipython-genutils   0.2.0
keyring            18.6.0
keyrings.alt       3.0.0
netifaces          0.10.4
numpy              1.19.5
pexpect           4.2.1
pickleshare        0.7.4
pip                9.0.1
pkginfo            1.6.2
prompt-toolkit     1.0.15
pycrypto           2.6.1
Pygments           2.2.0
pygobject          3.26.1
python-apt         1.6.5-ubuntu0.3
pytz               2021.1
pyxdg              0.25
PyYAML             3.12
requests           2.18.4
SecretStorage       2.3.1
setuptools          39.0.1
simplegeneric       0.8.1
six                1.11.0
soupsieve          2.3.2.2.post1
sqlparse           0.4.1
ssh-import-id      5.7
traitlets          4.3.2
typing-extensions  3.10.0.0
urllib3            1.22
wcwidth            0.1.7
wheel              0.30.0
```

Codio - Third-Party Packages

Guide

Collapse

Package Management

Installing Packages

The Python team maintains an official repository of third-party packages called the Python Package Index (PyPI). PyPI is pronounced "pie-pea-eye". The site has several hundred thousand packages. You can search the site for packages, and the index provides information about how to install the package, how to use it, its license, when it was last updated etc.

Once you know the package you want to install, you will use the tool `pip` to install the desired package. `pip` stands for "Python installer for packages". Before we begin installing anything, change to the `example1` directory and activate your virtual environment.

```
cd example1
source example-venv/bin/activate
```

Did you notice?

Once you see the `(example-venv)` in your prompt, you are ready to install a package. Let's start by installing `beautifulsoup4`, a popular package for scraping websites. In the terminal, enter the following command to install Beautiful Soup:

```
python3 -m pip install beautifulsoup4
```

Did you notice?

You should see a message in the terminal saying that Beautiful Soup was successfully installed. We can verify this using the `list` command. This will print out all of the installed packages.

```
python3 -m pip list
```

Some of the installed packages have titles that are not very helpful in understanding what they do. You can use the `show` command to get the metadata for a particular package.

```
python3 -m pip show soupsieve
```

You will see the following information:

```
Name: soupsieve
Version: 2.3.2.post1
Summary: A modern CSS selector implementation for BeautifulSoup.
Home-page: None
Author: None
Author-email: Isaac Muse <Isaac.Muse@gmail.com>
License: None
Location: /home/codio/workspace/example1/venv/lib/python3.6/site-packages
Requires:
```

Try these variations:

- In the file tree, open `example1`. Then open `example-venv`. Under `lib`, open `python3.6` and `site-packages`. You should see the Beautiful Soup package.
- Contrast your virtual environment with your Python installation. Deactivate `venv` and list all of the installed packages. Your virtual environment should have a different set of packages installed.

```
deactivate
python3 -m pip list
```

Installing Multiple Packages

If your virtual environment has been deactivated, be sure to activate it before continuing.

```
source example-venv/bin/activate
```

You can install multiple packages at once with `pip`. Let's say you are looking to get into scientific computing and data science. Two packages often used for these tasks are `scipy` and `pandas`. Just put a space between each package you want to install.

```
python3 -m pip install scipy pandas
```

These packages are a bit larger and have more dependencies, so the installation may take a little time.

Upgrading Packages

`pip` also allows you to upgrade installed packages. Use the normal command to install a package but add the `--upgrade` flag.

```
python3 -m pip install --upgrade beautifulsoup4
```

You can use the shortened flag `-u` instead of `--upgrade`. In addition, you can list out more than one package to upgrade.

```
python3 -m pip install -U scipy pandas
```

You can even upgrade `pip` itself.

```
python3 -m pip install -U pip
```

Reading Question

Assume you want to install the package `pybaseball`. What would the command be?

- ☐ `python3 install pybaseball`
- ☒ `pip3 install pybaseball`
- ☐ `pip3 uninstall pybaseball`
- ☐ `pip install pybaseball`

The correct answer is:

```
pip3 install pybaseball
```

- You should be using `pip3` so as not to use an old version of the program.
- `python3` is not a command to invoke the package manager.
- Use `install` not `uninstall` when trying to install a package.

Check it

Next